A Case Study Report

On

Design of a 3-Bit Synchronous Counter for Digital Timers

In the course

Digital Logic Design
(2501EC95)
In

B. Tech I Year I Sem

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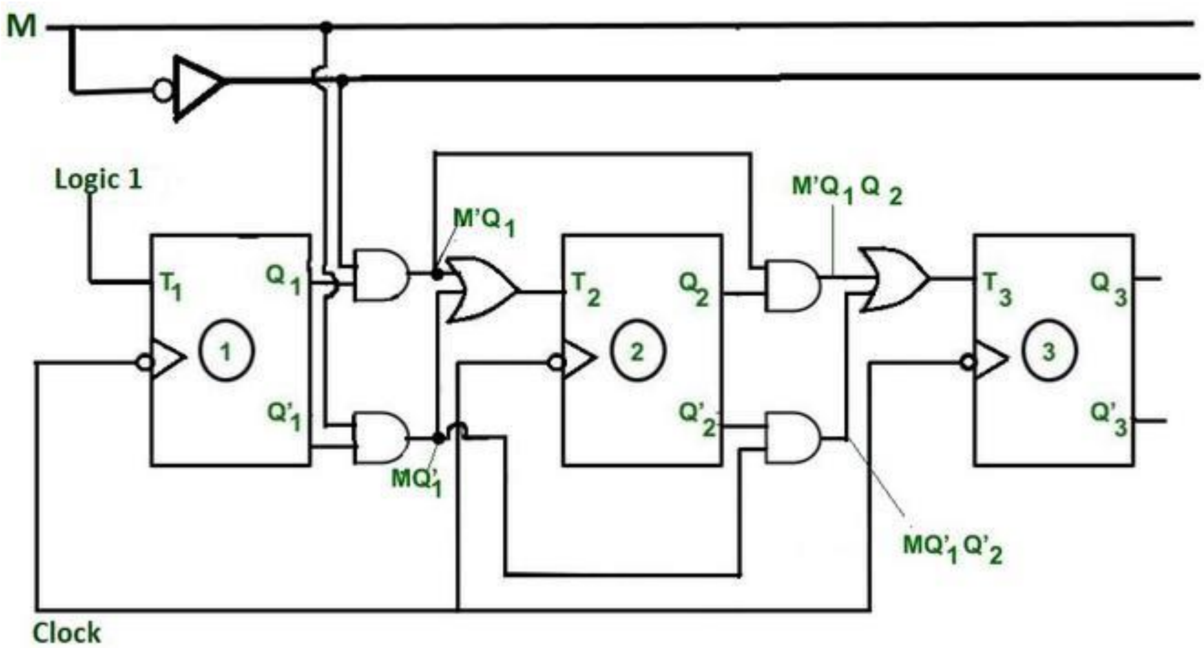
Faculty Signature

1. **Introduction**



Digital timers are widely used in electronic devices such as clocks, stopwatches, microwave ovens, and industrial controllers. A fundamental building block of these systems is a counter, which keeps track of time by counting clock pulses.
This case study focuses on the design of a 3-bit synchronous counter, emphasizing synchronous operation for accuracy, reliability, and ease of control.

2. Problem Statement

The objective of this case study is to design a 3-bit synchronous binary counter that:

- **Counts from 000 to 111 (0 to 7) in binary**
- **Updates all flip-flops simultaneously using a common clock**
- **Can be used as a basic module in digital timer applications**
- **Demonstrates predictable and stable operation**

2. **Background and Theory**

3.1 Counter

A counter is a sequential digital circuit that goes through a prescribed sequence of states upon the application of clock pulses.

3.2 Synchronous Counter

In a synchronous counter:

- All flip-flops are triggered by the same clock signal
- State changes occur simultaneously
- Delay problems common in asynchronous counters are avoided

3.3 Flip-Flop Selection

For this design, JK flip-flops are used due to their versatility and toggling capability.

4. Design Specifications

Parameter	Description
Counter Type	Synchronous Binary Counter
Number of Bits	3
Maximum Count	7 (111)
Flip-Flops Used	JK Flip-Flops
Clock	Common clock input
Application	Digital Timers

5. State Diagram

The counter follows the sequence:

000 → 001 → 010 → 011 → 100 → 101 → 110 → 111 → 000

This cyclic sequence repeats continuously with each clock pulse.

M	Q_3	Q_2	Q_1	Q_3^*	Q_2^*	Q_1^*	T_3	T_2	T_1
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	0	1	0	0	0	1

7. Logic Design

7.1 Flip-Flop Input Conditions

- **Flip-Flop Q0**
  - **Toggles at every clock pulse**
  - **J0 = 1, K0 = 1**
- **Flip-Flop Q1**

    - **Toggles when Q0 = 1**
    - **J1 = Q0, K1 = Q0**
- **Flip-Flop Q2**
    - **Toggles when Q0 AND Q1 = 1**
    - **J2 = Q0 · Q1, K2 = Q0 · Q1**

8. Block Diagram Description

The circuit consists of:

- **Three JK flip-flops connected in parallel to a common clock**
- **Logic gates (AND gates) to generate toggle conditions**
- **Outputs Q0, Q1, and Q2 forming the 3-bit binary count**

9. Application in Digital Timers

In digital timers:

- **The 3-bit synchronous counter can represent seconds or minutes**
- **It provides accurate timing due to synchronized operation**
- **Multiple counters can be cascaded to form higher-bit timers**

10. Advantages of Synchronous Counter

- **Reduced propagation delay**
- **Improved speed and reliability**
- **Easier timing analysis**
- **Suitable for high-frequency applications**

11. Conclusion

This case study presented the design and operation of a 3-bit synchronous counter used in digital timers. By employing synchronous design principles and JK flip-flops, the counter achieves reliable and predictable performance. Such counters serve as essential components in modern digital systems and timing applications.